BS2591 MACROECONOMIC THEORY AND APPLICATIONS

Tutorial 1
The demand side and the supply side

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LECTURE 1 · QUESTIONS 1 AND 2

Where to find the answers in the textbook

Q1 · Tobin's q

What does Tobin's q tell us firms' investment decisions depend upon? According to Tobin's q, when should a firm invest?

§1.2.6	Investment: Tobin's q theory, marginal q and average Q (pp. 45–49)
App. 1.5.3	Deriving Tobin's q: the condition q > 1 (pp. 57–59)
p. 60	Same as end-of-chapter Checklist question 9

Q2 · PIH and Tobin's q in the IS curve

Explain how the intuition behind (a) the PIH model and (b) Tobin's q theory is captured in the IS curve.

§1.2.3	The IS curve: $y = k(c_0 + a_0 + G) - ka_1 r$, the multiplier k and the investment function (pp. 30–32)
§1.2.5	Consumption: PIH, credit constraints and the size of the multiplier (pp. 33–45)
§1.3.2	Consumption, investment, and the IS curve (pp. 50–52)
p. 60	Same as end-of-chapter Problem 1

Carlin & Soskice (2024), Chapter 1. Page numbers are the printed book pages.

LECTURE 1 · QUESTION 3

Assuming the real interest rate is 4 per cent, calculate how, according to the PIH, consumption and borrowing would change in each of the following cases:

(a) A stock market crash permanently reduces the value of an individual's assets by 1,000.
(b) Households are told that in a year's time, they will receive a one-off bonus of 1,000. Then in one year's time, it is not paid.
(c) Comment briefly on your results.

Expected lifetime wealth (p. 35)

$$\Psi_t^E = (1+r)A_{t-1} + \sum_{i=0}^{\infty} \frac{y_{t+i}^E}{(1+r)^i}$$

assets carried in from last period + present value of expected income (sum over i = 0, 1, 2, ...)

PIH consumption function (p. 36)

$$C_t = \frac{r}{1+r}\,\Psi_t^E$$

with r = 0.04: r / (1 + r) = 0.04 / 1.04

Formulas: Carlin & Soskice (2024), Macroeconomics: Institutions, Instability, and Inequality, 2nd ed., OUP, pp. 35–36.

LECTURE 1 · Q3 (A)

A stock market crash permanently reduces assets by 1,000

Step 1 · The shock: assets fall

$$\Delta A_{t-1} = -1{,}000$$

Step 2 · Into lifetime wealth (income is unchanged)

$$\Delta \Psi^{E}_{t} = (1 + r)\, \Delta A_{t-1} = 1.04 \times (-1{,}000) = -1{,}040$$

Step 3 · Into the PIH consumption function

$$\Delta C_t = r / (1 + r) \cdot \Delta \Psi^{E}_{t} = (0.04 / 1.04) \times (-1{,}040) = -40$$

Direction: the minus sign. Consumption falls by 40 now and in every future period.

Borrowing: no change.

Formulas: Carlin & Soskice (2024), pp. 35–36.

LECTURE 1 · Q3 (B)

A promised bonus of 1,000 that is never paid

Step 1 · The news: lifetime wealth rises by the present value of the bonus

$$\Delta \Psi^{E}_{t} = 1{,}000 / (1 + r) = 961.54$$

Step 2 · Consumption rises at once, financed by borrowing

$$\Delta C_t = r / (1 + r) \cdot 1{,}000 / (1 + r) = (0.04 / 1.04^2) \times 1{,}000 = +36.98$$

Step 3 · The bonus is not paid: the loan must be repaid with interest

$$36.98 \times 1.04 = 38.46 = \text{fall in lifetime wealth}$$

Step 4 · Consumption falls, compared with before the news

$$\Delta C = -r / (1 + r) \times 38.46 = -1.48 \text{ for ever}$$

News: C rises by 36.98, by borrowing.
No bonus: C falls to 1.48 below its original level, the interest on the debt.

Formulas: Carlin & Soskice (2024), pp. 35–36.

LECTURE 1 · Q3 (C)

Comment: smoothing needs borrowing

Smoothing: Under the PIH, households smooth consumption over time

Credit constraint: The ability to borrow freely is crucial

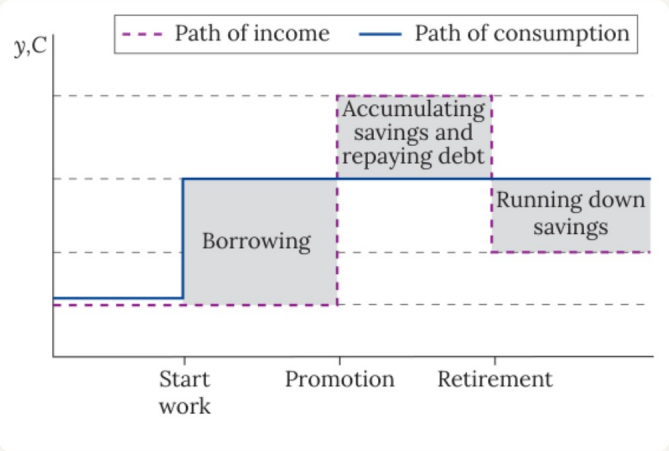


Figure 1.7 The PIH over the life cycle, p. 35

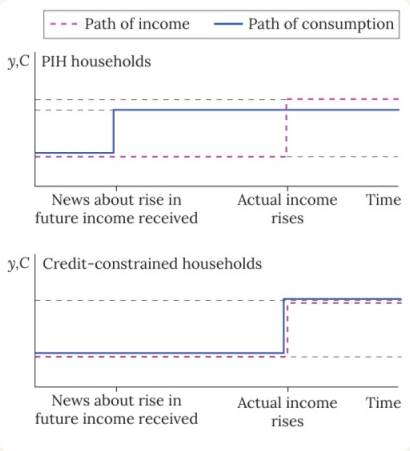


Figure 1.8, p. 40

Carlin, W. & Soskice, D. (2024) Macroeconomics: Institutions, Instability, and Inequality, 2nd ed., Oxford University Press.

LECTURE 1 · QUESTION 4

Use Section 1.3.2 to discuss what is expected to happen to the IS curve in response to the following shocks:

Shock	Reasoning	IS curve
(a) A crash in the stock market	Average Q: company values fall relative to replacement cost, so fixed investment falls.	Shifts left
(b) An increase in the retirement age	PIH: working longer raises permanent income, so households raise consumption now. But if it signals the state cannot afford pensions and health spending, consumption may not rise.	Shifts right (if no such signal)
(c) A decrease in the rate of depreciation	Marginal q: lower depreciation boosts investment.	Shifts right
(d) An increase in the cost of oil	For a net oil importer, expected future profits fall, so Tobin's Q falls.	Shifts left
(e) Faster technological progress	Marginal q: the marginal product of capital rises, so investment rises.	Shifts right

PART 1 · Q5 THE PARADOX OF THRIFT

If everyone saves more, does the economy save more?

Question: using the Keynesian cross, illustrate the paradox of thrift. Model the change in saving behaviour as an increase in the marginal propensity to save s_1 (with $c_1 + s_1 = 1$). Show how a rise in investment can offset the output fall associated with higher saving.

$$c1 + s1 = 1 \Rightarrow s1 \uparrow \Rightarrow c1 \downarrow$$

$$yD = c0 + c1(1 - t)y + I + G$$

$$\text{slope of AD} = c1(1 - t)$$

$$I = a0 - a1r$$

Since $c_1 + s_1 = 1$, a higher marginal propensity to save means a lower marginal propensity to consume.

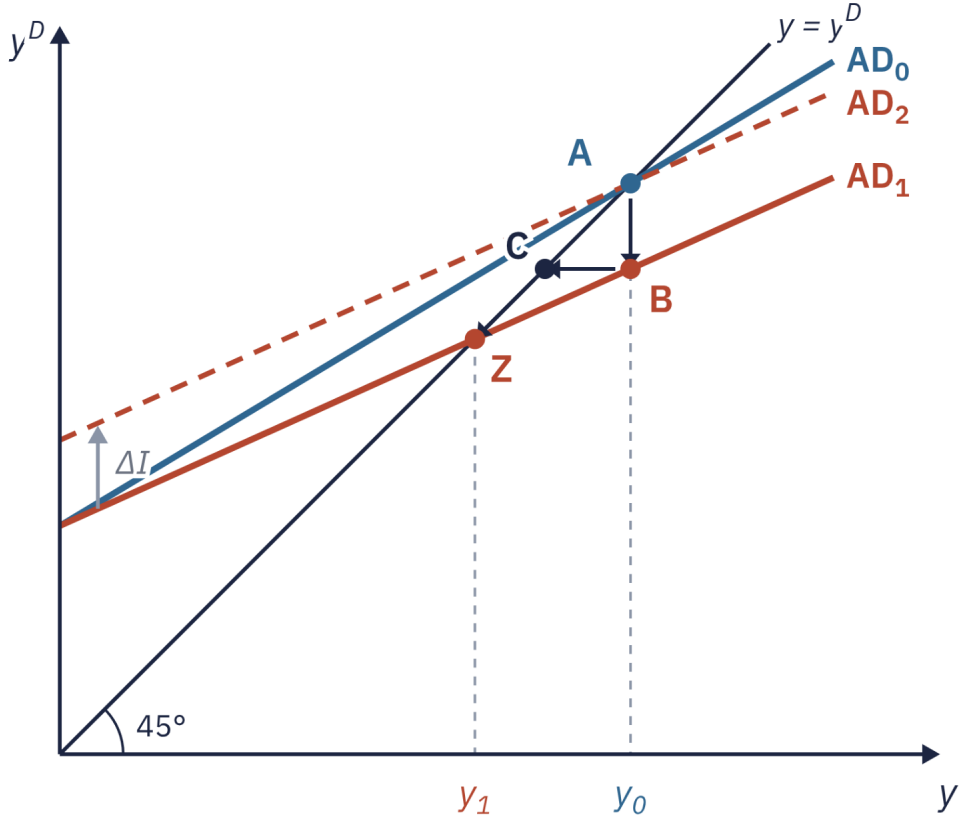
The slope of the AD line is $c_1(1 - t)$. When c_1 falls to c_1', the AD line becomes **flatter**, pivoting down around its intercept.

Investment depends on the interest rate, not on the saving rate, so the rise in s_1 has no effect on I.

Source: Tutorial 1 Q5 support notes, §1 and §3.

PART 1 · Q5 THE KEYNESIAN CROSS

A flatter AD line: from A to Z



AD0: $c0 + c1(1 - t)y + I + G$

AD1: $c0 + c1'(1 - t)y + I + G$

AD2 (dashed): $c0 + c1'(1 - t)y + I' + G$

A → B: households save more and consume less, so aggregate demand falls. At the old output y_0, output exceeds demand: $y > y^D$.

B → C: firms cut production, so output falls.

C → Z: lower output lowers income, which lowers consumption and demand again. This contractionary multiplier continues until the new equilibrium Z, with $y_1 < y_0$.

Figure: redrawn from Tutorial 1 Q5 support notes (answers Fig. 1.2). Text: §2.

PART 1 · Q5 WHAT HAPPENS TO SAVING

Saving is unchanged; only a rise in investment brings output back

What happens to saving

$$I = a0 - a1r, \quad S = I$$

Investment does not depend on the saving rate, so the increase in s_1 has no effect on investment. In equilibrium saving must equal investment, so **total saving is also unchanged**.

The higher propensity to save is exactly offset by the effect of lower income on saving.

Offsetting with higher investment

$$I \uparrow \Rightarrow y \uparrow \Rightarrow S \uparrow \text{ until } S = I'$$

A rise in investment $I \rightarrow I'$ shifts the flatter AD line up (AD_2, dashed). The increase in income raises saving until it equals the new, higher level of investment.

With a large enough rise in I, the economy returns to equilibrium at **A**.

The paradox: a policy that encourages saving in order to boost investment fails. Consumption, aggregate demand and output fall, while total saving and investment stay the same.

Source: Tutorial 1 Q5 support notes, §3 and §4.

PART 2 · CHAPTER 2

The supply side

LECTURE 2 · QUESTION 1

Why is the wage-setting curve upward sloping?

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Overview: Wage-Setting Curve

The wage-setting curve is upward sloping and above the reservation wage

- 'A': High U → high cost of job loss → only low w is required
- 'B': Low U → low cost of job loss → high w is required to induce worker effort

The wage-setting curve shifts up when:

- Unemployment benefit increases

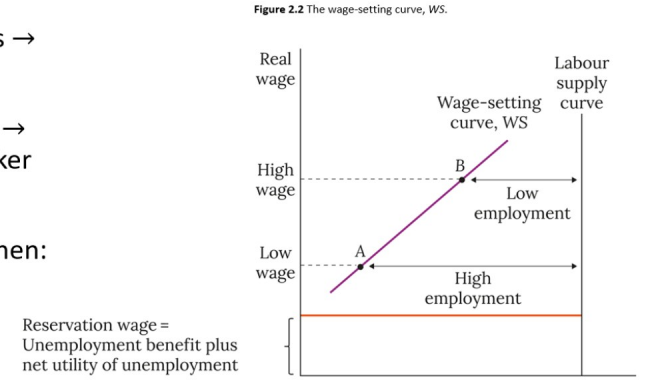


Figure 2.2 The wage-setting curve, *WS*.

The wage rises as unemployment falls (that is, as employment and output increase).

Effort

Boom periods will be associated with a higher probability of the worker getting another job, and thus the employer will have to offer a higher wage to secure adequate worker effort.

Recruitment and retention

A tighter labour market will also be associated with more difficulties in recruitment and retention leading the firm to set a higher wage.

Figure: Carlin & Soskice (2024), Figure 2.2 (Lecture 2 slides). Answer text: Lecture 2 tutorial answers, Q1.

LECTURE 2 · QUESTION 2

Explain in words and using a diagram what is meant by labour market clearing (refer to Appendix 2.5.1). Why is labour market clearing not observed in real-world labour markets?

Labour market clearing (Textbook Model)

- The labour market clears when the wage is such that there is neither excess supply of nor demand for labour.
- The market clears at the real wage where the labour demand curve intersects the labour supply curve: in a competitive labour market, at real wage w_0 and employment N^{cEcc}.

How the wage clears the market

If the wage is at $w_1 > w_0$
→ excess supply of labour: more people want jobs than firms want to hire
→ the unemployed offer to work for less, so the wage is bid down
→ back to w_0, where $N^D = N^s$

Anyone still without a job at w_0 does not want to work at that wage: unemployment U^{cEcc} is **voluntary**.

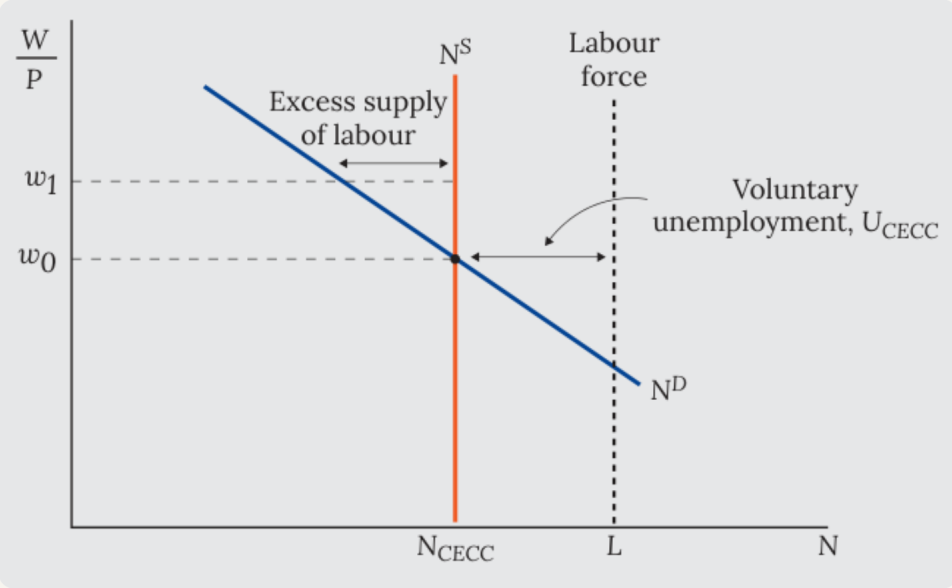


Figure 2.23 Market clearing, complete contracts (Appendix 2.5.1)

Figure: Carlin & Soskice (2024), Figure 2.23. Answer text: Lecture 2 tutorial answers, Q2.

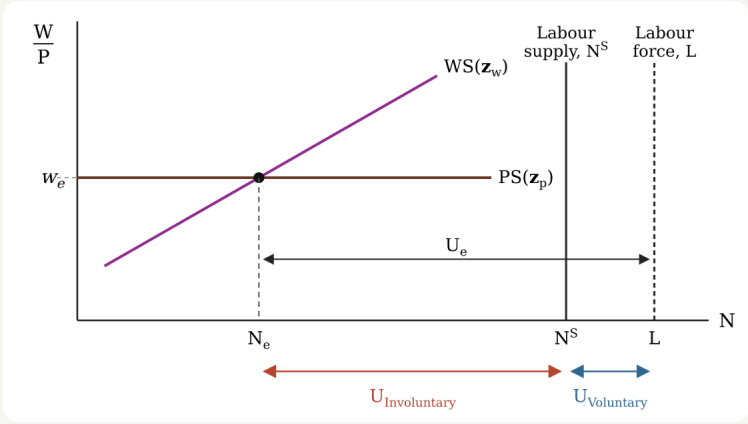
LECTURE 2 · QUESTION 2 (CONTINUED)

Why is labour market clearing not observed in real-world labour markets?

- Why would an employer not offer a job to a worker who is willing to work at a lower wage?

In the textbook model the employer would: the unemployed bid the wage down until the market clears at w_0.

- **Efficiency wage theory:** effort exerted by the employee is unobservable, so the employer sets the wage such that there is a cost to the worker of losing his job.
- The employer cannot specify in a contract every aspect of the work she requires, and is unable to perfectly monitor his output.
- This special feature of labour implies that labour market clearing is not observed in the data.



WS-PS equilibrium: involuntary and voluntary unemployment

How the wage is formed in WS-PS

Firms, not the market, set the wage: the no-shirking wage, higher when unemployment is low (WS) → firms' pricing fixes the real wage at $w = \lambda(1 - \mu)$ (PS) → equilibrium at $N_e < L$, with unemployment U_e → an offer to work for less than w_e is not credible (the worker would shirk), so the wage does not fall

U_e is **involuntary**: the unemployed would take a job at w_e, but there are only N_e jobs.

Figure: adapted from Carlin & Soskice (2024), Figures 2.11 and 2.14. Answer text: Lecture 2 tutorial answers, Q2.

LECTURE 2 · QUESTION 3

Derive the price-setting curve. What does the equation for the PS curve tell us about the ability of firms to make supernormal profits?

A firm maximises profits when marginal revenue (*MR*) equals marginal cost (*MC*).

• If labour is the only factor of production, then:

$$MC = \frac{W}{MPL}$$

where *W* is the nominal wage and *MPL* is the marginal product of labour.

• Marginal revenue is:

$$MR = \frac{dR}{dy} = P + y\frac{dP}{dy}$$

Defining the price elasticity of demand as

$$\eta = -\frac{dy}{dP} \cdot \frac{P}{y}$$

we can rewrite *MR* as:

$$MR = P\left(1 - \frac{1}{\eta}\right)$$

• Profit maximisation requires *MR* = *MC*, so:

$$P\left(1 - \frac{1}{\eta}\right) = \frac{W}{MPL}$$

• Rearranging gives the price-setting equation:

$$P = (1+\mu)\frac{W}{MPL}, \qquad \mu = \frac{1}{\eta - 1}$$

where μ is the mark-up on marginal labour costs.

• The ability of firms to make supernormal profits is linked to the mark-up μ. Under perfect competition, demand is perfectly elastic ($\eta \to \infty$), so μ = 0.

LECTURE 2 · QUESTION 4

The equation you have derived shows that firms set prices as a mark-up over unit labour costs. Use this equation to show the relationship between wage growth and price growth in the economy. What if mark-ups were not constant, but depended positively on inflation?

From the price-setting equation

$$P = (1+\mu)\frac{W}{\lambda}$$

the price growth equation is:

$$\left(\frac{\Delta P}{P}\right)_t = \left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta \lambda}{\lambda}\right)_t$$

price inflation = wage inflation – productivity growth

Since productivity is assumed to be constant in the baseline model,

$$\left(\frac{\Delta \lambda}{\lambda}\right)_t = 0$$

then wage changes will translate one-for-one into price changes:

$$\left(\frac{\Delta P}{P}\right)_t = \left(\frac{\Delta W}{W}\right)_t$$

In the baseline model firms immediately adjust prices after any wage round, in order to protect their profit margins, so prices move 1-to-1 with wages.

If μ rises when inflation rises

$$\mu = \mu(\pi), \qquad \mu'(\pi) > 0$$

then price dynamics pick up an additional term from changes in μ.

Using the same price-setting relationship:

$$P_t = (1+\mu_t)\frac{W_t}{\lambda_t}$$

Taking growth rates (approximately, in plain terms):

$$\left(\frac{\Delta P}{P}\right)_t \approx \left(\frac{\Delta W}{W}\right)_t - \left(\frac{\Delta \lambda}{\lambda}\right)_t + \Delta \ln(1+\mu_t)$$

$$\pi_t \approx \omega_t - g_{\lambda,t} + \Delta \ln(1+\mu_t)$$

where:

$$\pi_t = \left(\frac{\Delta P}{P}\right)_t$$

$$\omega_t = \left(\frac{\Delta W}{W}\right)_t$$

$$g_{\lambda,t} = \left(\frac{\Delta \lambda}{\lambda}\right)_t$$

$\Delta \ln(1+\mu_t)$ is the change in the log mark-up term.

Inflation rises

↓

Firms raise mark-ups: $\Delta \ln(1+\mu_t) > 0$

↓

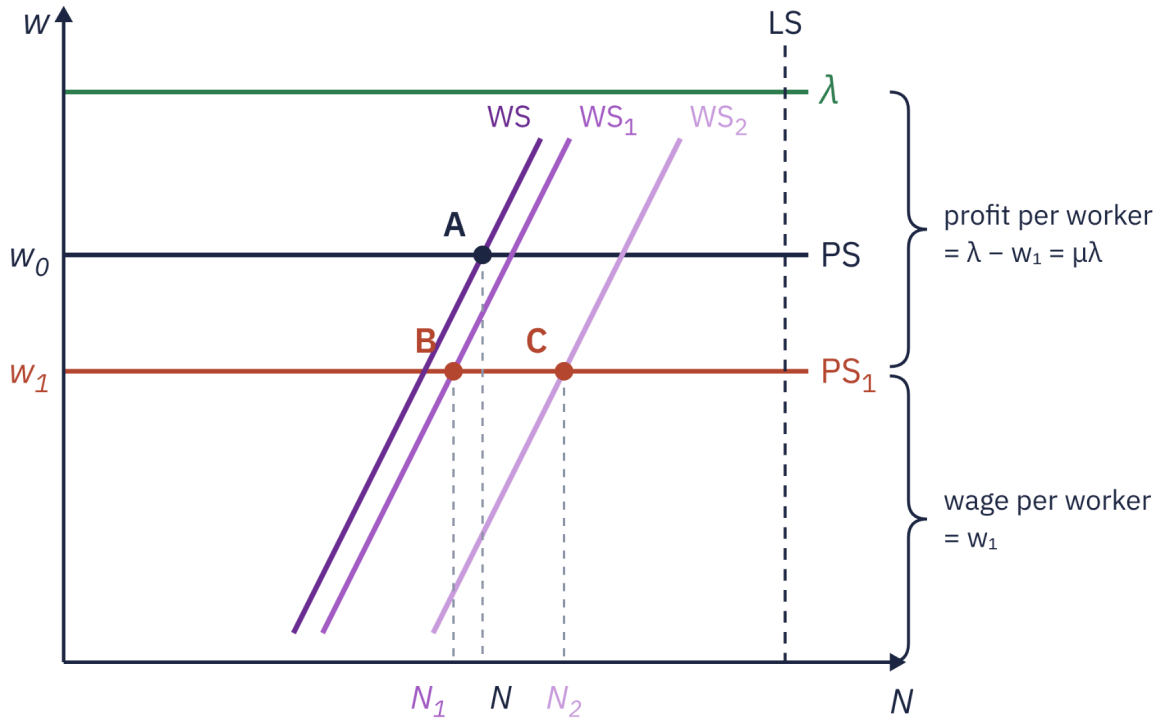
Prices rise even faster than wages: π_t responds more than one-for-one to $\omega_t - g\lambda,_t$

↓

Pass-through is amplified: additional inflationary pressure

LECTURE 2 · QUESTION 5 · WS-PS

More firm power, weaker unions: what happens to wages and jobs?



- **Firm power** ↑ ⇒ less competition ⇒ μ ↑ ⇒ PS shifts down to PS_1.
- **Union strength** ↓ ⇒ wage-setting power ↓ ⇒ WS shifts right, to WS_1 or WS_2.

Results

(a) The real wage falls unambiguously: $w_0 \to w_1$, since $w = \lambda(1 - \mu)$ and μ ↑.

(b) Employment is ambiguous: it depends on the relative size of the two shocks.

- WS_1 (point B): $N_1 < N$, employment falls.
- WS_2 (point C): $N_2 > N$, employment rises.

μ here is the markup on price, i.e. the profit share.

LECTURE 2 · QUESTION 5 · LORENZ CURVE

What happens to inequality?

Three groups: the **unemployed** (income 0), **workers** (wage share 1 – μ), **owners** (profit share μ).

- **Distribution channel:** μ′ > μ, so the kink moves down and the curve moves away from equality (area A, +).
- **Unemployment channel:** the share of the population with zero income changes.

Unemployment rises (WS₁, $u' > u$): both channels push the curve away from equality, so inequality **rises unambiguously** (as in Fig. 2.19).

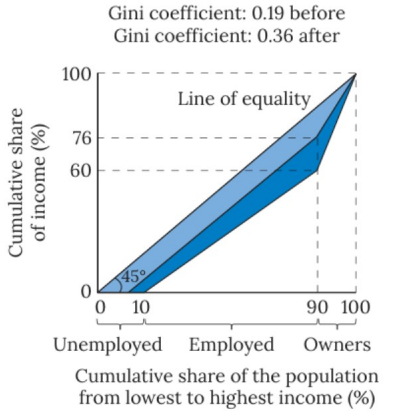
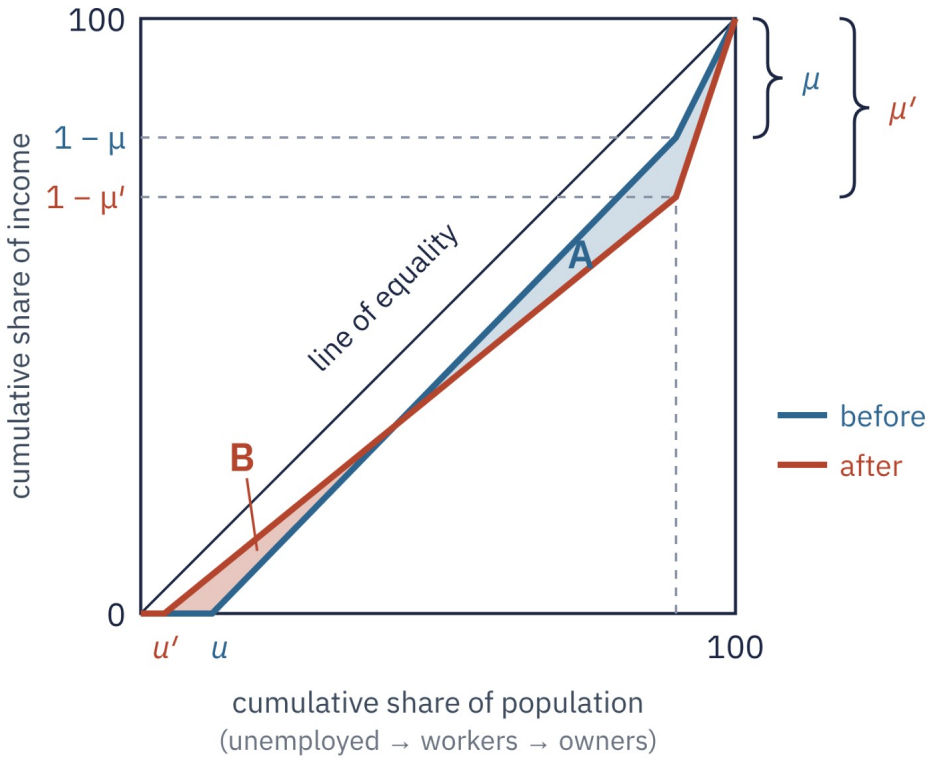


Fig. 2.19, C&S (2024), p. 101



Unemployment falls (WS₂, $u' < u$, as drawn): fewer people have zero income, so the bottom of the curve moves towards equality (area B, –).

A > B: inequality rises · A < B: falls · A = B: no change

Questions?

Email me any time: **MUB@CARDIFF.AC.UK**

Next week · Chapter 3: the 3-equation model